Credit Card Fraud Detection with XGBoost

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Problem and summary of the Data

Problem:

Predict whether a credit card transaction is fraudulent or not

Data Overview:

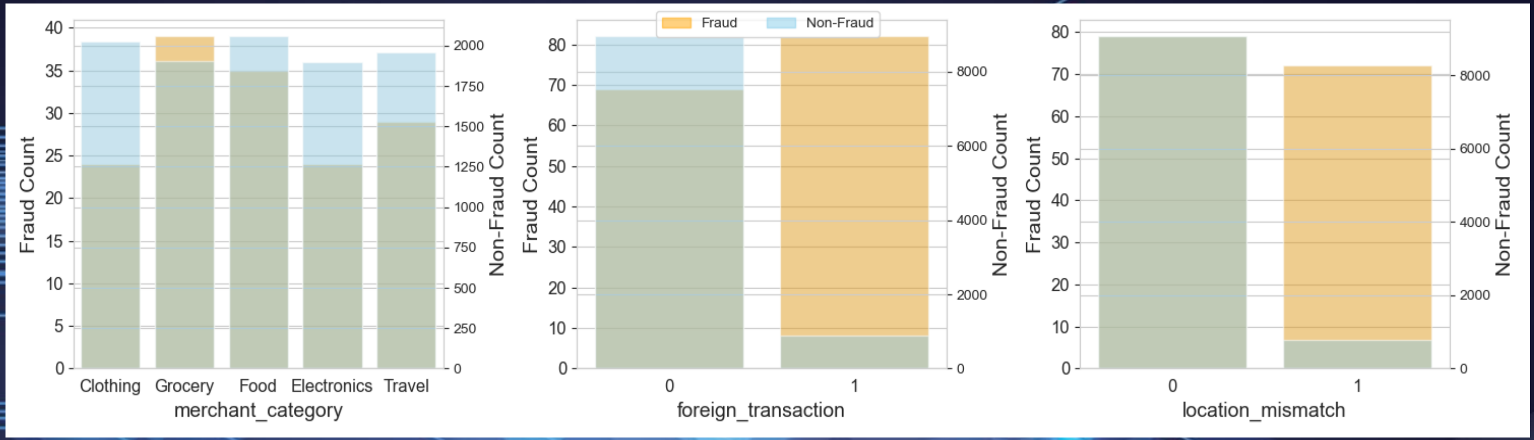
- 10,000 credit card transactions.
- 10 features including transaction_id, amount, transaction_hour, foreign_transaction, location mismatch, etc.
- Target variable: is_fraud = 0 or 1.
- Highly unbalanced dataset.
  - 94.49% non-fraudulent transactions
  - 1.51% fraudulent transactions.

Exploratory Data Analysis (EDA)

Transaction density for every hour

- More fraudulent transactions are taking place during the early hours.
- Non-fraudulent transactions are evenly distributed across the hours.

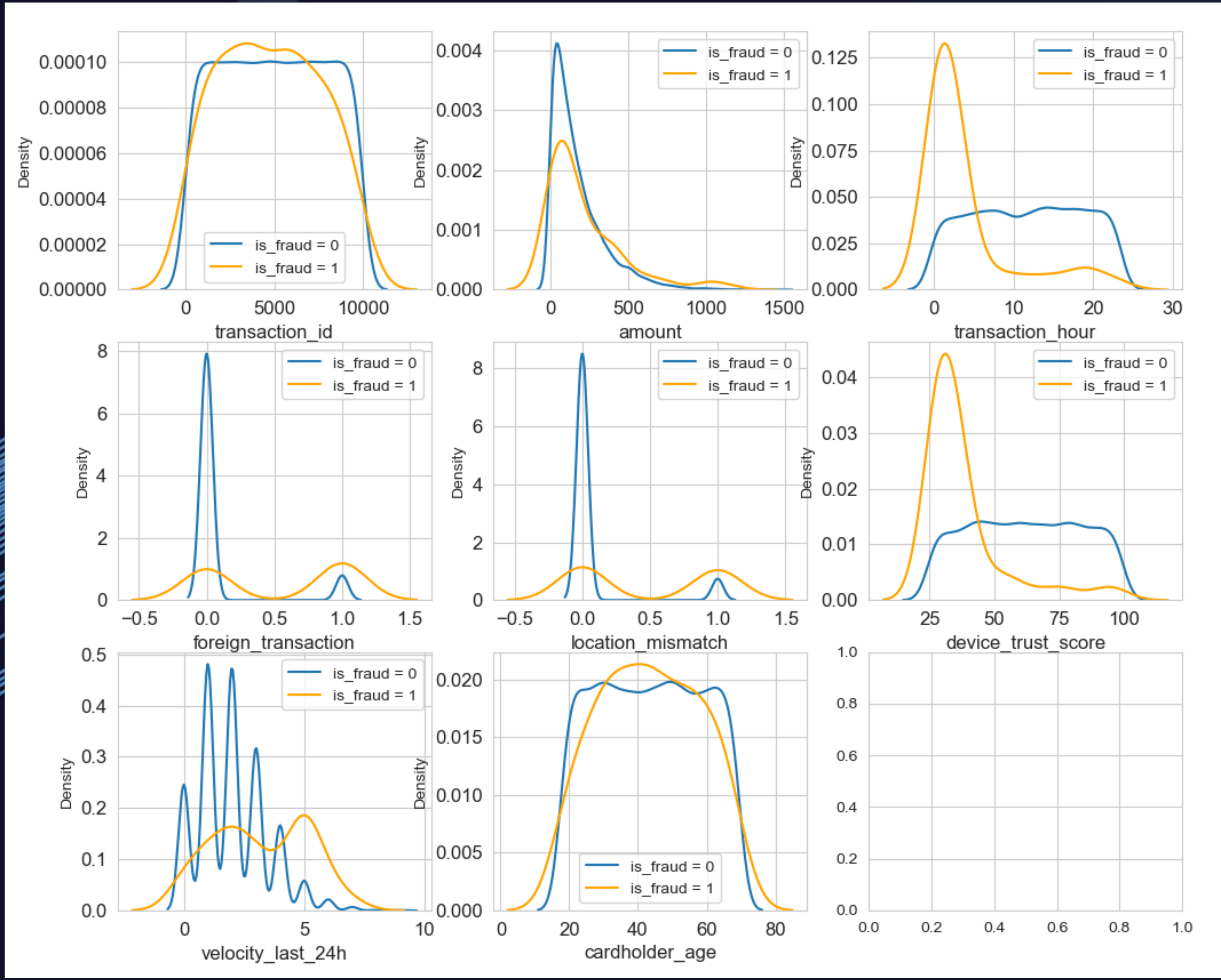
Categorical Features



- Fraud counts are lower than the non-fraud counts as expected.
- **Merchant category** -> doesn't show a significant variation in fraud activities based on the category.
- **Foreign transaction** -> Fraud counts are similar for both foreign and local transactions. Non-fraud counts follow a different trend
- **Location mismatch** -> Fraud counts are similar for both cases of location mismatch and match. Non-Fraud counts is high for location_mismatch =0 and low for location_mismatch =1

Location mismatch and Foreign transactions seem to be good features to classify fraud from non-fraud transactions

Exploratory Data Analysis (EDA)



- The following features appears to be good predictors to classify fraudulent activities from non-fraudulent activities.
  Transaction_hour,
  device_trust_score,
  location_mismatch,
  foreign_transaction,
  velocity_last_24h

Building model using XGBoost

XGBoost (eXtreme Gradient Boosting):

- Fast and efficient
- Excellent for imbalanced datasets which is the case here
- Built-in regularization to avoid overfitting

Accuracy metric used: **ROC-AUC** (Area under the Receiver Operating Characteristic Curve) which is excellent for testing accuracy for imbalanced datasets.

model Test AUC score: 99.99%

Cross validation score:

obtained a cross-val score of 99.98%

Final Results:

- Model built using XGBoost can classify non-fraudulent from fraudulent transactions with an accuracy metric (AUC score) of 0.9998.
- Based on the Feature Importance, location_mismatch, velocity_last_24h, foreign_transaction are effective predictors in classifying.